Intimate Partner Violence & Weight Gain in Young Women

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INTRODUCTION

One in four women in the US are affected by **intimate partner violence (IPV).**[1]

IPV is associated with **obesity-related diseases.** [2]

In a nationwide study, IPV **exposure during adolescence** was associated with subsequent weight gain.[3]

- An additional study examined the association between IPV exposure and weight gain in Egyptian women. No literature examined **exposure during young adulthood** and subsequent weight gain in American women.
- Understanding the associations between IPV and weight will inform **effective interventions** in preventing risk of obesity-related disease among women with IPV histories.
- We investigated associations between **physical and sexual IPV exposure** and subsequent weight gain in women over **5 years**.

STUDY QUESTION

Are young women with IPV histories at a higher risk for subsequent weight gain than women without IPV histories?

METHODS

Longitudinal Study

Data: *Project EAT survey*; middle and high school students in the Minneapolis-St. Paul area aged 11-18 (*1998-1999*)

Exposures: IPV measured using *Project EAT survey*

Outcome: *Weight gain*, based on change in BMI (*2004-2009*)

EAT I: 1998

EAT II: 2004

EAT III: 2009

Data Analysis:
Linear regression of 5-year BMI change on IPV exposure and covariates.

RESULTS

Baseline covariates (sociodemographics) according to type of IPV

Figure 1. Race: Distribution according to type of IPV exposure

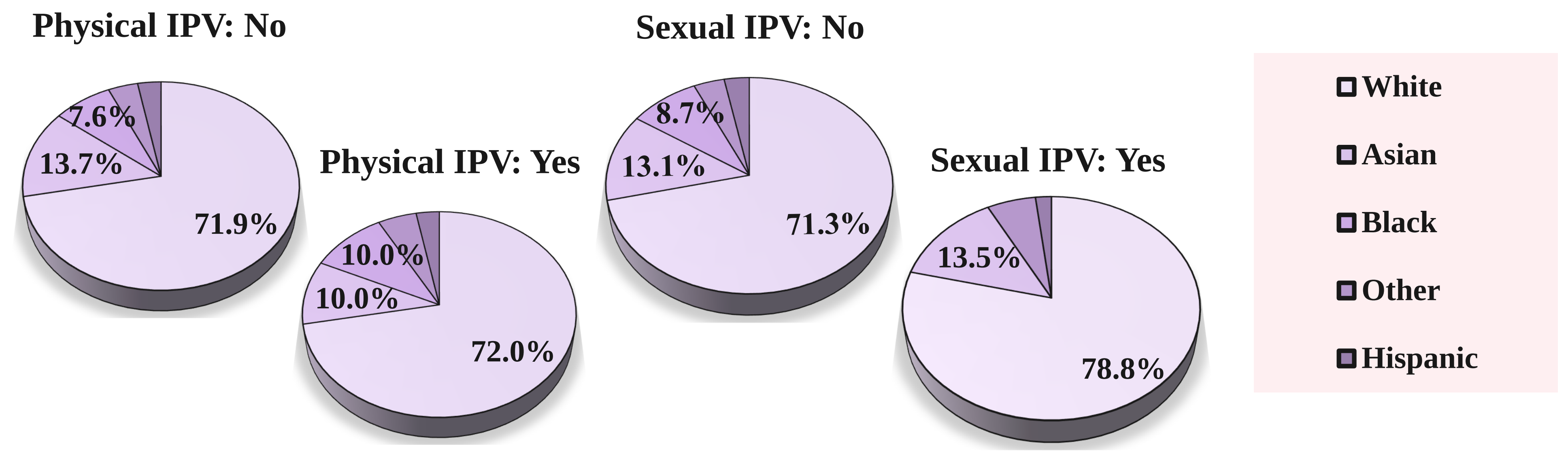
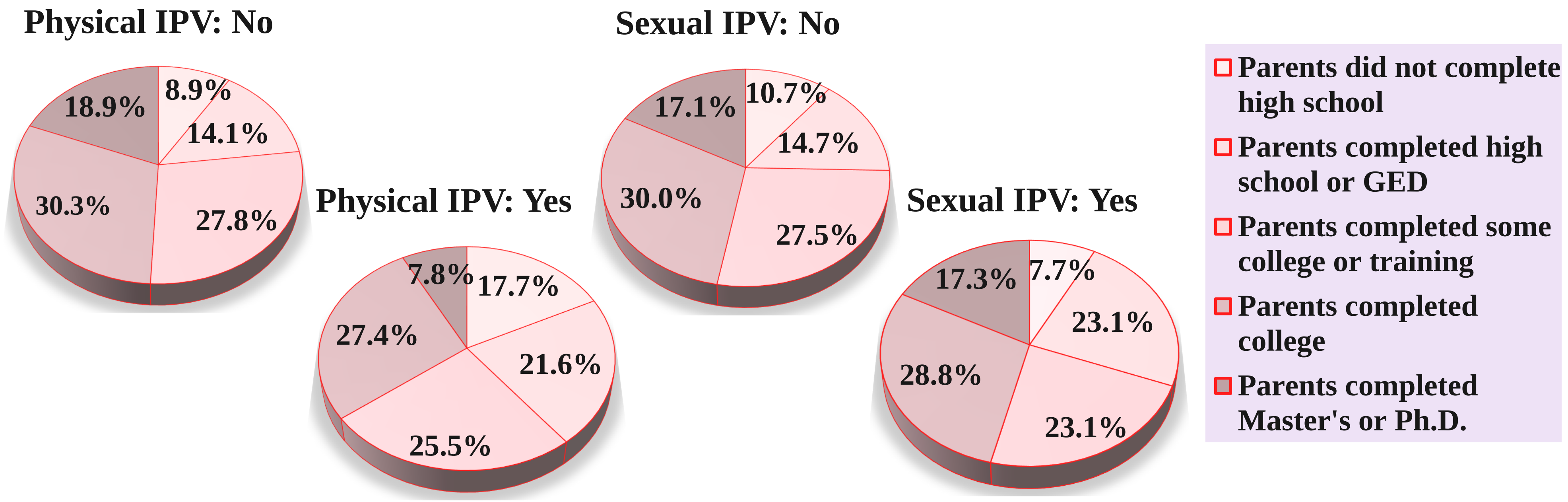


Figure 2. Socio-economic Status: Distribution according to type of IPV exposure



DISCUSSION

- Findings suggest there is **a positive association** between physical and/or sexual IPV exposure and subsequent 5-year BMI change in **women.**
- BMI increases were associated with exposure to both sexual and physical IPV, even **after adjustment** for **sociodemographics** and **baseline BMI.**
- The study demonstrates a **longitudinal relationship** between IPV exposure in **young adults** (18-23) and BMI change.
- IPV, in our study, was ascertained prior to weight change allowing for **identification of the temporal order** from IPV to weight.
- Findings were consistent with a hypothesized **causal mechanism.**

LIMITATIONS & STRENGTHS

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- Single-item questions rather than validated scales for IPV measurements
- Potential for response bias and under-reporting

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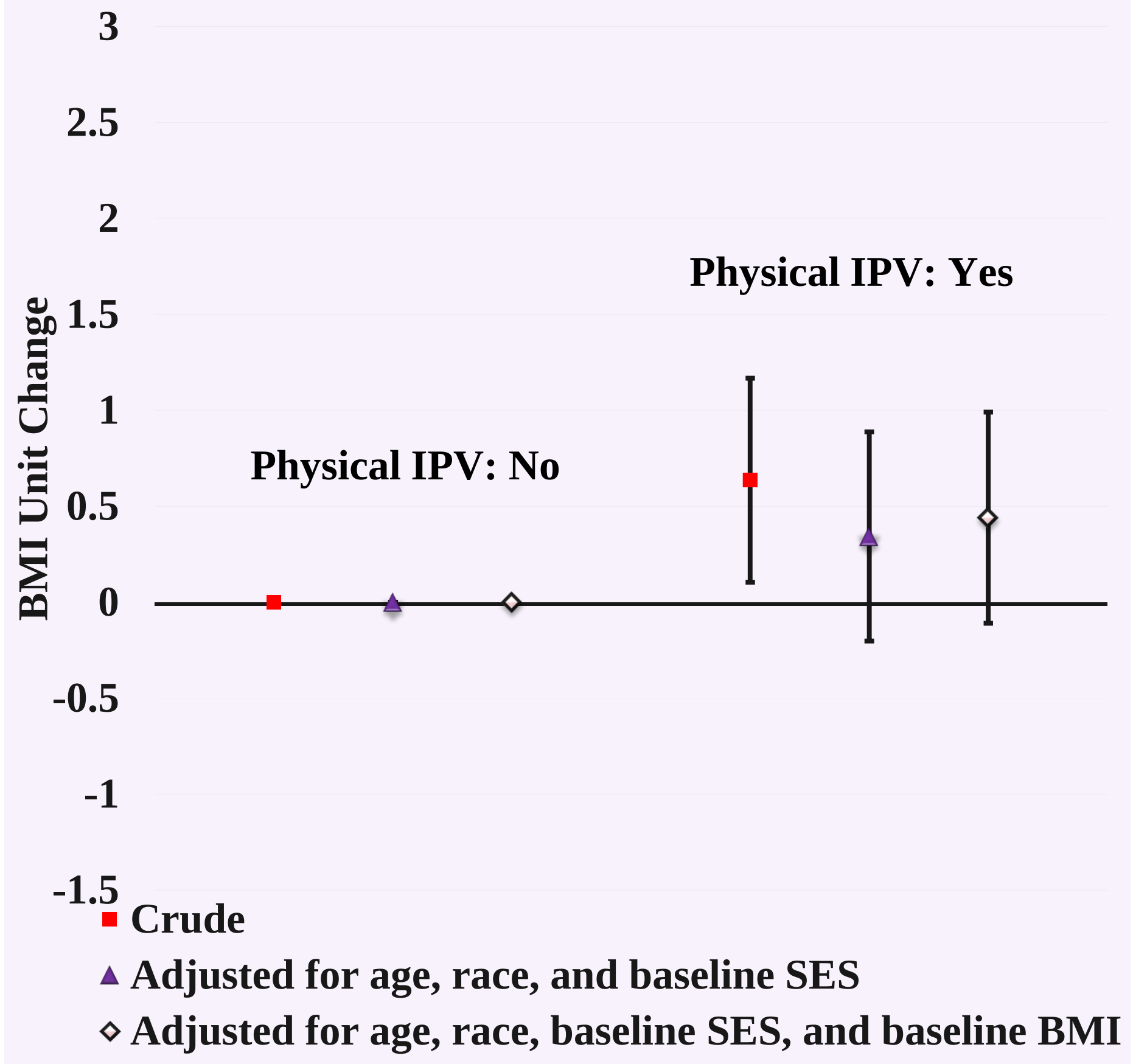
+ Similar prevalence of IPV as representative US samples
+ Relatively large sample
+ Prospectively measured IPV and well-validated self-reported BMI measures

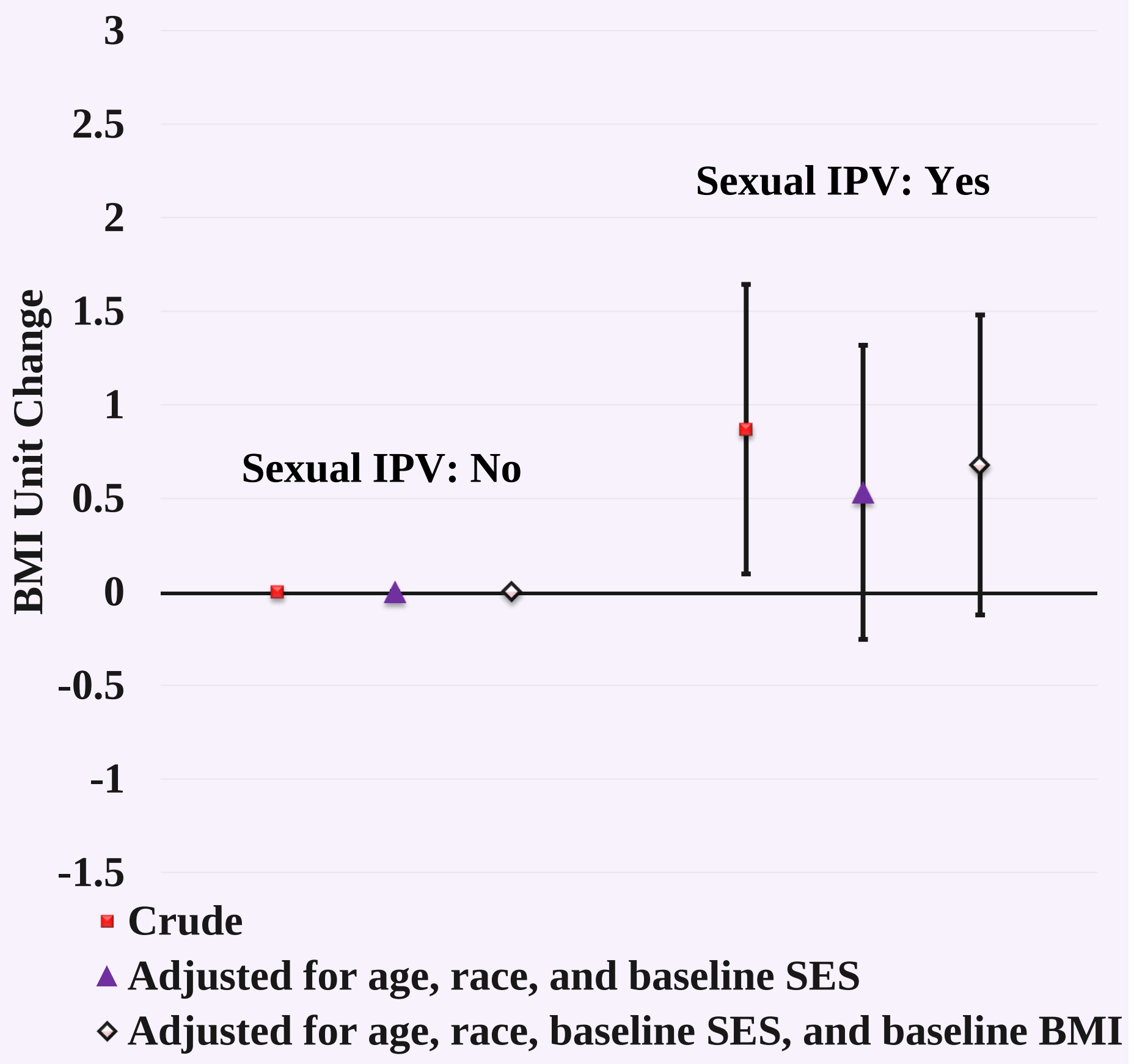
RESULTS CONTINUED

Figure 3. Estimated difference in 5-year BMI change associated with physical IPV



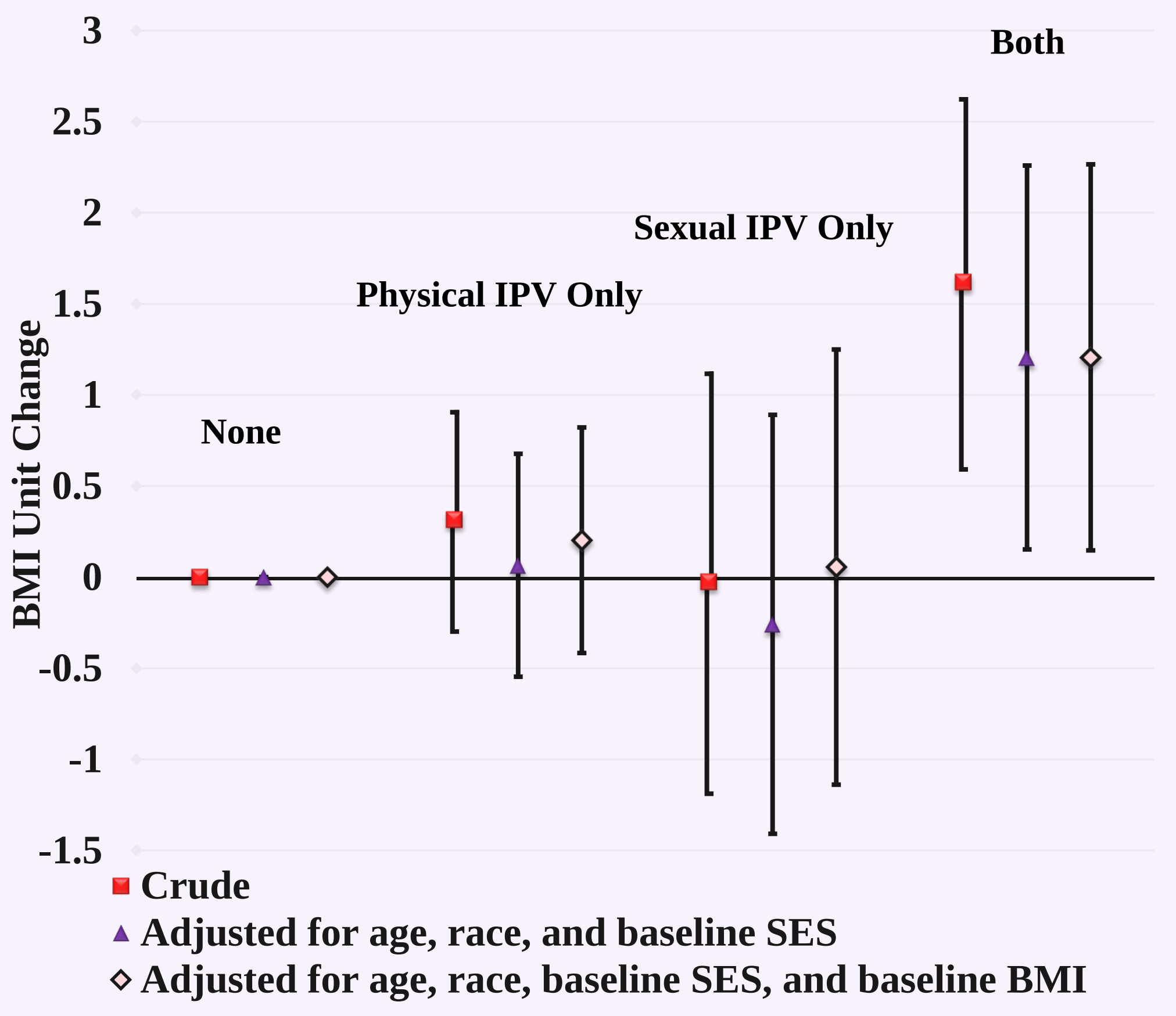
Number of participants who stated—NO: 530 | YES: 103

Figure 4. Estimated difference in 5-year BMI change associated with sexual IPV



Number of participants who stated—NO: 580 | YES: 53

Figure 5. Estimated difference in 5-year BMI change associated with physical IPV only, sexual IPV only, and both



Number of participants who stated—None: 509 | Physical IPV Only: 71 | Sexual IPV Only: 21 | Both: 32

Notable Trends:

- **Figure 1:** Greater proportion of women reporting sexual IPV exposure **identified as white** when compared to women without sexual IPV exposure.
- **Figure 2:** Women reporting physical IPV had **lower** parental education completion.
- **Figures 3, 4, 5:** Association between weight change and IPV was **attenuated, but remained positive** when adjusted for covariates and BMI particularly for women who reported both physical and sexual IPV.

IMPLICATIONS

- Future studies need to examine the affects of both severity and frequency of different forms of IPV on weight longitudinally in men and women from different cultural groups and during different life stages.
- Clinicians working with patients struggling with obesity should be aware that a history of IPV may play a role in weight. Development of tailored counseling approaches that address the specific needs of patients with histories of IPV is warranted.
- Policy makers need to incorporate programs in schools to educate students regarding the signs of IPV and its impacts on physical and psychological health.

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3. Clark CJ, Spencer RA, Everson-Rose SA, Brady SS, Mason SM, Connett JE, Henderson KM, To M, Suglia SF. Dating violence, childhood maltreatment, and BMI from adolescence to young adulthood. Pediatrics. 2014; 134(4):678-85.

Photo 1. Web: http://www.vov.com/issues/preventing-intimate-partner-violence, Photo 2. Web: http://images.idiva.com/media/content/2013/Oct/weightgain1.jpg

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